Phase 8 | Project Demonstration

Scalability & Future Plan

AI-Powered Travel Planner | Smart Situational Itinerary Engine

Live URL:	https://travel-planner-iota-plum.vercel.
GitHub:	github.com/itskian2007/travel-planner
Tech:	React + FastAPI + PostgreSQL + Vercel
Team:	itskian2007

1. Current Architecture Scalability

The Travel Planner is built on a cloud-native, horizontally scalable architecture. Vercel's edge network auto-scales the frontend globally. The FastAPI backend is designed for async operation, supporting 1,000+ concurrent connections. PostgreSQL supports read replicas for horizontal database scaling.

2. Current Scalability Benchmarks

Frontend (Vercel)		
- Auto-scales to unlimited	- 0ms cold start	- 99.99% uptime SLA

Backend (FastAPI)		
- 1,000+ concurrent users	- <500ms p95 response	- Async ASGI server

Database (PostgreSQL)		
- 10,000 rows/sec writes	- Connection pooling	- Read replica ready

Image API (Wikipedia)		
- 100 req/min rate limit	- Response cache layer	- Fallback strategy

3. Product Roadmap

v2.0 - Q1 2025

- Hotel & flight booking API integration
- PDF itinerary export functionality
- Social sharing: share itineraries
- User reviews & destination ratings
- Real-time weather integration
- Push notifications for travel reminders

v3.0 - Q3 2025

- AI/ML personalisation (collaborative filtering)

- Natural language itinerary requests

- Multi-destination trip chaining

- Offline mode with service worker caching

- Multi-language support (10+ languages)

- AR destination preview via camera

v4.0 - Q1 2026

- Mobile native apps (React Native)
- Real-time collaborative trip planning
- Travel budget tracker + currency conversion
- Travel agency white-label partnerships
- Expansion to 100+ destinations globally
- AI concierge chatbot for in-trip assistance

4. Scalability Architecture Plan

Proposed architecture upgrade for 100K+ users:

Component	Technology	Scaling Strategy
Load Balancer	NGINX / AWS ALB	Distribute traffic across multiple backend instances
Backend Scaling	Kubernetes (k8s)	Auto-scale FastAPI pods based on CPU/memory load
Database Scaling	PostgreSQL + Redis	Read replicas + Redis caching for frequent queries
CDN	CloudFlare Enterprise	Global image and asset delivery with DDoS protection
Monitoring	Prometheus + Grafana	Real-time metrics, alerting, SLA dashboards
CI/CD	GitHub Actions + Vercel	Automated testing and deployment pipeline
Search	Elasticsearch	Full-text destination and attraction search at scale
Image Storage	AWS S3 + CloudFront	Scalable image storage replacing Wikipedia API dependency